MILESTONE 2

HR AGENT

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BACKGROUND & OBJECTIVES

CHALLENGE

Employees need reliable HR policies, their own leave balances, and clear guidance on leave requests.

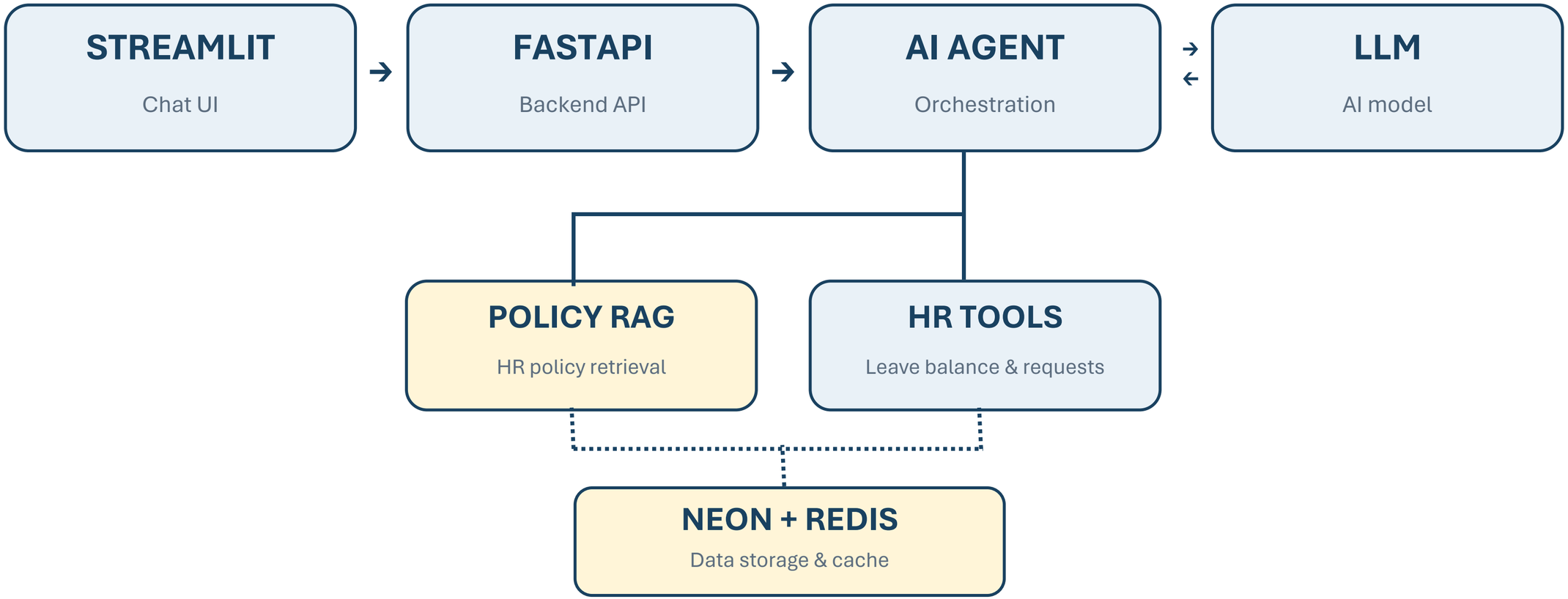
SOLUTION

An HR agent combines LLM reasoning, policy retrieval, authenticated employee data, and workflow tools.

OBJECTIVE

Give evidence-based answers, protect employee data, and keep approval under human control.

SYSTEM ARCHITECTURE & KEY FEATURES



KEY FEATURES • **HR policy Q&A with citations** • **Check leave balance** • **Submit leave request**

QA RESULTS 15 SCENARIOS

9

PASS · 60%

QA-04–10, QA-13 and QA-15 met visible response criteria.

6

NOT PASS · 40%

QA-01, QA-02, QA-03, QA-11, QA-12, QA-14

DEEP-DIVE SCOPE

QA-01 / QA-03 / QA-11 / QA-12 / QA-14 + QA-02 unsolicited-ticket safety finding

QA-01 | NOT PASS

MISSING POLICY CITATION

QUESTION

Jatah cuti tahunan 12 hari, tetapi berapa saldo saya sekarang?

EXPECTED

The agent should show Alya’s 8-day balance and explain the 12-day annual entitlement with a policy citation.

WHAT THE AGENT DID

The agent showed both numbers, but did not include a citation for the leave policy.

TRACE EVIDENCE

Only get_my_leave_balance was called. search_hr_policy was not called.

WHAT WENT WRONG

The agent treated a question about both balance and policy as a balance-only question.

PROPOSED IMPROVEMENT

Improve _needs_policy_evidence() in agent.py so mixed questions use both tools and cite the policy.

QA-02 | NOT PASS

UNREQUESTED HR TICKET

QUESTION

Saya ingin membawa 8 hari cuti ke tahun depan. Bisa?

EXPECTED

Explain the 5-day carry-over limit and the March 31 deadline. Do not create a ticket unless the user asks.

WHAT THE AGENT DID

The policy answer included a citation, but the agent also said it created an HR ticket.

TRACE EVIDENCE

search_hr_policy (ok) → create_hr_ticket (ok). The user did not ask to create a ticket.

WHAT WENT WRONG

The backend allowed a write tool without checking explicit user authorization. The ticket record was not independently verified.

PROPOSED IMPROVEMENT

Add a consent check before create_hr_ticket. Do not offer write tools for information-only questions.

QA-03 | NOT PASS

INCORRECT POLICY ABSTENTION

QUESTION

Saya sakit tiga hari. Boleh saya kirim diagnosis di chat ini?

EXPECTED

Explain sick-leave requirements and direct medical details to the official HR channel.

WHAT THE AGENT DID

The agent said there was not enough policy evidence and did not answer the user’s question.

TRACE EVIDENCE

search_hr_policy succeeded and retrieved HR-LEAVE-002 and HR-PRIV-001; the result was abstained.

WHAT WE KNOW / DO NOT KNOW

Relevant documents were found, but we do not yet know why the model or validation checks rejected the answer.

PROPOSED IMPROVEMENT

Log the reason for abstention, investigate the exact cause, then fix it without removing privacy checks.

QA-11 | NOT PASS

GENERIC PROPOSAL ERROR

QUESTION

Siapkan cuti tahunan mulai besok.

EXPECTED

Explain the minimum three-working-day notice rule and avoid creating an invalid leave proposal.

WHAT THE AGENT DID

The agent only said the request could not be processed, without explaining why.

TRACE EVIDENCE

Balance (ok) → policy (ok) → prepare_leave_request (error).

WHAT WE KNOW / DO NOT KNOW

A tool error became a generic message. The original exception was not recorded, so the exact cause is still unknown.

PROPOSED IMPROVEMENT

Use structured business errors and show clear explanations for leave rules, while hiding internal system errors.

QA-12 | NOT PASS

WEEKEND-ONLY LEAVE

QUESTION

Siapkan cuti tahunan hanya Sabtu dan Minggu.

EXPECTED

Explain that a Saturday–Sunday-only request contains zero working days and ask for dates if needed.

WHAT THE AGENT DID

The agent asked for specific dates, but did not clearly explain the zero-working-day rule.

TRACE EVIDENCE

Only search_hr_policy was called. prepare_leave_request was not called.

WHAT WENT WRONG

The existing business_days() validation was not reached through the proposal flow.

PROPOSED IMPROVEMENT

Explain the weekend rule clearly, request exact dates, and validate concrete ranges with business_days().

QA-14 | NOT PASS

MISSING REQUEST DATES

QUESTION

Saya mau cuti tahunan, tolong langsung ajukan.

EXPECTED

Ask for the start date and end date before preparing or submitting any leave request.

WHAT THE AGENT DID

The agent said policy evidence was insufficient instead of asking for the missing dates.

TRACE EVIDENCE

search_hr_policy (ok) → abstained. No early date clarification occurred.

WHAT WENT WRONG

The agent missed the submission intent because its intent check relies on narrow keywords.

PROPOSED IMPROVEMENT

Expand leave-request intent detection in agent.py, then require both dates before proceeding.

IMPROVEMENT ROADMAP & NEXT PHASE

Prioritize safe actions, precise error handling, and regression coverage before deployment.

1 P0 · ACTION SAFETY

First, I will add a backend consent check. The agent must not create HR tickets unless the user has approved the action.

2 P1 · ANSWER QUALITY

Next, I will improve policy citations for mixed questions. For QA-03, I will log why the agent abstains before deciding what to fix.

3 P1 · LEAVE WORKFLOW

I will make leave errors easier to understand, explain why weekend-only leave has zero working days, and ask for missing dates.

NEXT PHASE — UI REDESIGN & HOSTING

After these fixes, I will improve the Streamlit UI, rerun all 15 QA tests, and prepare secure deployment of Streamlit and FastAPI with Neon and Redis.



THANK YOU